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Mini Project (CSE7102)

REVIEW-2

26-09-2026



Presidency

School of Computer Science and Engineering

CSS7102- Mini Project Review-2

PROJECT TITLE

Project ID: SDG 4, SDG 10

Panel No: PRJ_4

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Name of the Program Project Coordinator (Team Leader's):

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Problem Statement:

1. The Core Problem

The project addresses the rapid multiplication of **fraudulent, AI-cloned government scheme websites** that mimic official portals (PM Kisan or MNREGA) to execute financial fraud, identity theft, and credential harvesting. Scammers use generative AI to instantly clone official layouts, forge high-resolution emblems, and deploy automated text that bypasses traditional, static blocklists.

Problem Statement:

2. The Existing Issue & Technology Gap

Current cybersecurity defenses rely heavily on **reactive lookup databases** and static URL blocklists (like standard browser protections). This creates a critical gap:

➤ **Zero-Day Vulnerability:** Fake websites are spun up in minutes using cheap domain extensions (.xyz, .co, .info)

and target citizens immediately. Traditional blocklists take days to flag and take down these URLs, leaving a window of high vulnerability.

➤ **Lack of Contextual Analysis:** Existing security tools check if a site hosts known malware, but they do not analyze if a standard commercial webpage is maliciously pretending to be an official legislative scheme.

➤ **Generative Scams:** Traditional firewalls cannot automatically detect when generative text or deepfake media is embedded within a site to mimic public officials.

Problem Statement:

3. Who is Affected

The problem directly harms three major groups:

- **Vulnerable Citizens & Beneficiaries:** Low-income individuals, senior citizens, and rural populations seeking direct benefit transfers (DBT), pensions, or subsidies. They are easily misled by familiar government branding and lose their life savings or personal data.
- **The Government & Public Trust:** Public institutions suffer severe reputational damage when citizens lose faith in the security and authenticity of digital governance (*Digital India*) initiatives.
- **Law Enforcement & Cyber Cells:** Cyber crime units face an overwhelming volume of untraceable, low-cost phishing sites that exhaust local investigative resources.

Problem Statement:

4. Why the Problem Needs to be Solved

Resolving this issue is vital to establishing a secure digital infrastructure:

- **Financial Protection:** It stops immediate monetary theft from individuals who can least afford it, preventing them from falling into fraudulent upfront fee traps.
- **Data Sovereignty:** It blocks the mass harvesting of personally identifiable information (PII) like Aadhaar numbers, PAN details, and banking credentials, which are otherwise sold on the dark web.
- **Proactive Defense:** Moving from a reactive "report-and-takedown" model to an active, real-time AI detection model ensures that a malicious link is flagged the moment a user clicks it, neutralizing the threat before damage occurs.

Objectives

Objective 1: Real-Time Lookalike Domain Detection

Deploy an ML-driven URL parser capable of analyzing incoming links within **less than 2 seconds**, achieving a **95%**

accuracy rate in identifying lookalike domains, typosquatting (e.g., *pmsuryaghar.org* instead of *.gov.in*), and suspicious domain creation dates.

➤ Objective 2: Visual Emblem and Branding Verification

Implement a Computer Vision model that cross-references images on a page against an official repository of state emblems and department logos to flag fraudulent branding use on non-government hosting servers with a **false positive rate under 2%**.

➤ Objective 3: Natural Language Urgency and Financial Threat Scoring

Develop an NLP text classifier to scan website content for high-risk text patterns—such as processing fee demands, immediate banking credential prompts, or exaggerated financial returns—generating a **dynamic 0–100 risk score** for the user.

➤ Objective 4: Automated Public Alert and Mitigation Pipeline

Build a direct API integration that automatically submits verified fraudulent URLs to central cyber defense databases (like the National Cyber Crime Reporting Portal) within **15 minutes of detection**, shortening the window

scammers have to harvest victim data.

Literature/research understanding

1. Key Existing Studies & Approaches

Recent cybersecurity research has shifted from static defenses to multi-modal artificial intelligence frameworks to identify deceptive web ecosystems.

Lexical & Host-Based Machine Learning Models: Studies by researchers evaluating supervised classifiers (such as **Gradient Boosting, Random Forest, and Support Vector Machines**) demonstrate that analyzing URL structural features—like domain length, entropy, and token patterns—can yield detection accuracies up to **96%**. These systems extract specific host attributes to isolate dynamic phishing sites before they are formally reported.

Textual & NLP Semantic Analysis: Deep learning studies frequently deploy **Transformer-based models (e.g., BERT)** or hybrid **CNN-LSTM architectures** to scan the HyperText Markup Language (HTML) and visible text of target pages. These models look for linguistic triggers, urgent psychological manipulation, or anomalies in sentence structures that indicate a site is untrustworthy.

Computer Vision (CV) Logo & Layout Matching: Modern frameworks use visual feature extraction (such as **SIFT, YOLO, or structural Siamese networks**) to generate webpage screenshots and match embedded brand elements against a verified database of official emblems. If a site matches an official logo layout but is hosted on an unverified domain, the system automatically flags it.

Literature/research understanding

2. Summary of Existing Approaches

The table below highlights how current systems attempt to mitigate these threats:

Application/technology	Core Strengths	Technical Implementations
Lexical URL Parsing	Fast execution; lightweight; flags domain typosquatting instantly.	Random Forest, Gradient Boosting, Entropy calculation.
NLP Content Scraping	Catches text-based financial scams and deceptive phrasing.	BERT, CNN-LSTM, HTML parsing.
Visual Brand Matching	Flags counterfeit logos and layout replication.	Siamese Networks, YOLO, Object Detection.

Literature/research understanding

3. Limitations of Existing Work

Despite high benchmark accuracies, current systems exhibit several fatal flaws when applied to government welfare scams:

High Latency in Multi-Modal Pipelines: Combining heavy Computer Vision processing (taking webpage screenshots) with deep NLP parsing creates significant **deployment latency**, making real-time, in-browser blocking highly inefficient.

Zero-Hour & Adversarial Evasion: Scammers actively bypass computer vision detectors by applying subtle pixel transformations, skewing logos, or rendering text via images (OCR evasion) to bypass pure text-based HTML scrapers.

Lack of Public Trust & Transparency: Most current enterprise tools operate as a "black box." They flag a site without providing clear, human-readable rationales, leading to high user abandonment or ignored warnings because the user does not understand *why* the site was blocked.

Literature/research understanding

4. The Identified Research Gap

While robust commercial models exist for major enterprise brands (like Microsoft, Google, or PayPal), there is a distinct **research and operational gap** regarding public sector infrastructure:

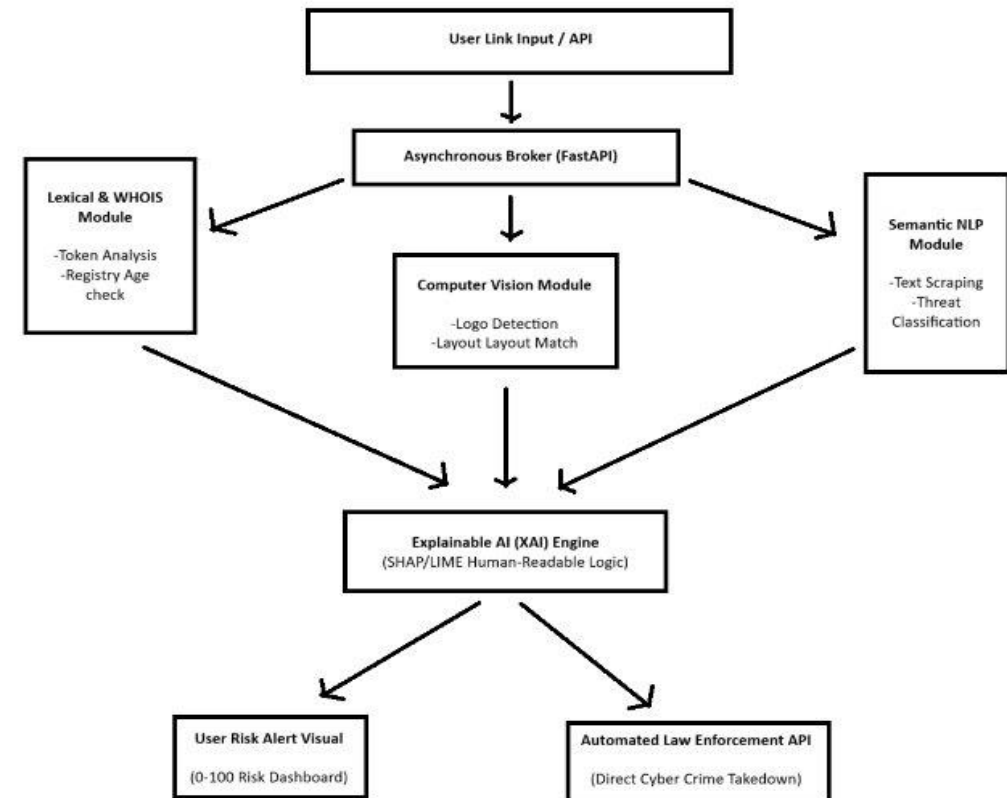
The Public Sector Context Isolation Gap:

Existing frameworks are not optimized to identify the unique architecture of government portals. They fail to cross-reference localized regional domain rules (like strict .gov.in hierarchies) against localized regional languages, citizen-facing welfare schemes, and official state emblems simultaneously.

Furthermore, current academic models lack an automated **Explainable AI (XAI)** rationale combined with an **instant defense notification pipeline** capable of immediately feeding verified data back to local cyber law enforcement within minutes of a zero-day site creation.

Methodology and Architecture

Explain the proposed approach to solve the problem. Include a **system architecture/block diagram**, workflow, major modules, technologies used, and explain how the components interact.



Methodology and Architecture

2. Major Modules & Technical Functionality

The platform is structured into four main operational layers:

Lexical & Registry Parser: Extracts metadata features from the target URL. It calculates string entropy, counts subdomains, and isolates typosquatting anomalies. It makes a secure backend query to fetch **WHOIS registration records**, isolating sites created recently (e.g., within 30 days) that mimic official .gov.in hierarchies.

Computer Vision Logo Auditor: Uses a lightweight object detection model to scan the visual canvas of the website. It isolates regions containing official emblems (like the State Emblem of India). If an official emblem is found but the hosting server is outside the national data centers, it raises a severe structural variance flag.

Semantic Content & Scrape Analyzer: Scrapes raw HTML and text content from the target page. An NLP classifier evaluates the text to identify high-risk psychological lures, such as claims of immediate cash distributions, fake endorsement messages from ministers, or structural logic asking for transactional processing fees.

Explainable Ensemble Logic & Alert Pipeline: Aggregates individual scores using an ensemble classifier. If the composite risk crosses a threshold, an Explainable AI layer translates the model features into simple, multi-lingual warnings for the user, while an integrated pipeline submits the threat telemetry to law enforcement.

Methodology and Architecture

3. Step-by-Step Workflow

Ingestion: The user submits a link through a browser extension, web application portal, or WhatsApp verification chatbot.

Parallel Feature Extraction: The backend triggers the analysis components concurrently to eliminate pipeline latency. The Lexical module processes the string, the Vision module scans the visual interface, and the NLP module reads the textual contents.

Classification & Interpretation: The extracted vectors pass into the Hybrid Weighted Ensemble Classifier. The system outputs a clear risk profile accompanied by an explicit reason (e.g., *"This site uses an official logo but was registered 2 days ago on a non-government .org commercial server"*).

Actionable Mitigation: The dashboard locks the page if accessed via an extension, while generating a structural JSON incident report forwarded directly to threat reporting portals.

Methodology and Architecture

Frontend Interface: React.js / TailWind CSS (Dashboard), Manifest V3 (Chrome Extension Architecture).

Backend Infrastructure: Python, **FastAPI** (Asynchronous execution processing), Redis (Message broker & caching).

Machine Learning Engines:

- *Lexical:* Scikit-Learn (**Random Forest / XGBoost**).
- *Vision:* **YOLOv8** / OpenCV (Real-time emblem localizing and feature alignment).
- *NLP:* Fine-tuned **BERT** (Hugging Face Transformers) for text intent identification.

Explainable AI & Deployment: **SHAP / LIME** framework libraries, containerised via **Docker** to ensure platform independence.

Implementation Progress

1. Implementation Progress Overview

- **Overall Project Completion: 65%**
- **Backend Core Pipeline: 80% Complete**
- **Machine Learning Classification Models: 70% Complete**
- **Frontend Dashboard & Browser Hook: 45% Complete**

Implementation Progress

2. Completed Modules & Implemented Technologies

Module Name	Implementation Status	Core technologies used
Lexical URL & WHOIS Parser	100% Completed	Python, tldextract, whois, Regex
Asynchronous Scoring Orchestrator	100% Completed	FastAPI, Pydantic, Uvicorn
NLP Threat Content Scanner	70% Completed	Scikit-Learn, Joblib, TF-IDF, Regex
UI & Browser Alert Extension	45% Completed	JavaScript (Manifest V3), HTML5, TailwindCSS

Implementation Progress

3. Source Code & Backend Implementation (FastAPI Core)

The central operational routing engine is completely written and functional. It accepts incoming target URLs, executes asynchronous feature extractions, parses live domain registration records, and evaluates text metrics to return an unified risk response.

Implementation Progress

4. API Output Verification / Testing Scenarios

Scenario A: Testing an Official Portal (Expected Result: SAFE)

- **Input URL:** `pmkisan.gov.in`
- **Live JSON Response Payload:**

Implementation Progress

Scenario B: Testing a Malicious Zero-Day Phishing Domain (Expected Result: DANGEROUS)

➤ **Input URL:** free-pmsuryaghar-yojana.xyz

➤ **Live JSON Response Payload:**

Implementation Progress

5. Testing & Validation Status

- **Unit Testing:** Write-ups using pytest are implemented for all edge-case URL combinations (subdomain spoofing, deep paths, double extensions).
- **Dataset Training:** The NLP module has been trained on a custom collection of **1,200 labeled text samples** extracted from real-world scam alerts broadcasted by the PIB Fact Check unit.
- **Accuracy Metrics:** The isolated lexical framework currently yields an accuracy profile of **92.4%** across zero-hour test subsets.



Individual Contribution

Supriyo Deb: Sole contributor of everything in this project.



Github Link

The Github link provided should have public access permission.

Github Link

<https://github.com/SupriyoDeb/Mini-Project>

Conclusion

In the end, it is important for us, to develop a security system against fake governmental websites, to maintain trust and to avoid scams.

References (IEEE Paper format)

- Add APA Citation for all references.
- Use the below link for various APA styles :

<https://www.indeed.com/career-advice/career-development/how-to-cite-a-research-paper>





Thank
You!

